

DevelopersHub Corporation

AI/ML Engineering Internship Tasks

Due Date: 26th June 2025

Overview

As part of your AI/ML Engineering Internship at DevelopersHub Corporation, you are required to complete **at least 3 out of the following 6 tasks**. These tasks are designed to help you build a strong foundation in artificial intelligence and machine learning through real-world datasets and use cases.

You will practice skills like data preprocessing, model training, evaluation, prompt engineering, and even chatbot development — all of which are essential in AI/ML careers.

Task 1: Exploring and Visualizing a Simple Dataset

Objective:

Learn how to load, inspect, and visualize a dataset to understand data trends and distributions.

Dataset:

Iris Dataset (CSV format, can be loaded via seaborn or downloaded)

Instructions:

- Load the dataset using `pandas`.
- Print the shape, column names, and the first few rows using `.head()`.
- Use `.info()` and `.describe()` for summary statistics.
- Visualize the dataset:
 - Create a scatter plot to show relationships between features.

- Use histograms to show value distributions.
- Use box plots to identify outliers.
- Use `matplotlib` and `seaborn` for plotting.

Skills:

- Data loading and inspection using pandas
 - Descriptive statistics and data exploration
 - Basic plotting and visualization with seaborn and matplotlib
-

Task 2: Predict Future Stock Prices (Short-Term)

Objective:

Use historical stock data to predict the next day's closing price.

Dataset:

Stock market data from Yahoo Finance (retrieved using the `yfinance` Python library)

Instructions:

- Select a stock (e.g., Apple, Tesla).
- Load historical data using the `yfinance` library.
- Use features like Open, High, Low, and Volume to predict the next Close price.
- Train a Linear Regression or Random Forest model.
- Plot actual vs predicted closing prices for comparison.

Skills:

- Time series data handling
- Regression modeling

- Data fetching using APIs ([yfinance](#))
 - Plotting predictions vs real data
-

Task 3: Heart Disease Prediction

Objective:

Build a model to predict whether a person is at risk of heart disease based on their health data.

Dataset:

Heart Disease UCI Dataset (available on Kaggle)

Instructions:

- Clean the dataset (handle missing values if any).
- Perform Exploratory Data Analysis (EDA) to understand trends.
- Train a classification model (Logistic Regression or Decision Tree).
- Evaluate using metrics: accuracy, ROC curve, and confusion matrix.
- Highlight important features affecting prediction.

Skills:

- Binary classification
 - Medical data understanding and interpretation
 - Model evaluation using ROC-AUC and confusion matrix
 - Feature importance analysis
-

Task 4: General Health Query Chatbot (Prompt Engineering Based)

Objective:

Create a chatbot that can answer general health-related questions using an LLM (Large Language Model).

Tools:

OpenAI GPT-3.5 via API (or use a free open-source model like Mistral-7B-Instruct on Hugging Face)

Instructions:

- Build a simple Python script or notebook that sends user queries to an LLM.
- Use prompt engineering to make the responses friendly and clear (e.g., "Act like a helpful medical assistant").
- Add safety filters to avoid giving medical advice that could be harmful.
- Example queries:
 - "What causes a sore throat?"
 - "Is paracetamol safe for children?"

Skills:

- Prompt design and testing
 - Using APIs for LLMs (e.g., OpenAI, Hugging Face)
 - Safety handling in chatbot responses
 - Building simple conversational agents
-

Task 5: Mental Health Support Chatbot (Fine-Tuned)

Objective:

Build a basic chatbot that provides supportive and empathetic responses for stress, anxiety, and emotional wellness.

Model Base:

DistilGPT2, GPT-Neo, or Mistral (7B)

Dataset for Fine-Tuning:

EmpatheticDialogues (Facebook AI)

Instructions:

- Fine-tune a small LLM using Hugging Face's Trainer API.
- Train it to respond empathetically using real human dialogues.
- Ensure the tone is gentle and emotionally supportive.
- Build a command-line or Streamlit-based interface to test it.

Skills:

- Model fine-tuning with Hugging Face Transformers
 - Emotional tone design for safe chatbots
 - Conversation modeling
 - Deployment using CLI or simple web apps
-

Task 6: House Price Prediction

Objective:

Predict house prices using property features such as size, bedrooms, and location.

Dataset:

House Price Prediction Dataset (available on Kaggle)

Instructions:

- Perform preprocessing on features like square footage, number of bedrooms, and location.
- Train a regression model (Linear Regression or Gradient Boosting).
- Visualize predicted prices compared to actual prices.

- Evaluate with Mean Absolute Error (MAE) and RMSE.

Skills:

- Regression modeling
 - Feature scaling and selection
 - Model evaluation (MAE, RMSE)
 - Real estate data understanding
-

Submission Requirements (Checklist for Each Task)

To receive credit for a task, ensure the following:

1. Jupyter Notebook

- Clear problem statement and goal
- Dataset loading and preprocessing
- Data visualization and exploration
- Model training and evaluation
- Explanation of results and final insights

2. Code Quality

- Code should be clean, modular, and commented
- Include explanations of each major step

3. GitHub Repository

- Create a GitHub repo for your AI/ML internship tasks

- Add a **README.md** file summarizing:
 - Task objective
 - Dataset used
 - Models applied
 - Key results and findings

4. Submission on Google Classroom

- Share your GitHub repository link for each completed task.
-

Important Note

- You are required to complete **at least 3 out of the 6 tasks** before the due date: **26th June 2025**.
 - You may complete more tasks for additional learning and project portfolio strength.
 - If you need help, feel free to ask questions in the group or consult with your mentors.
-